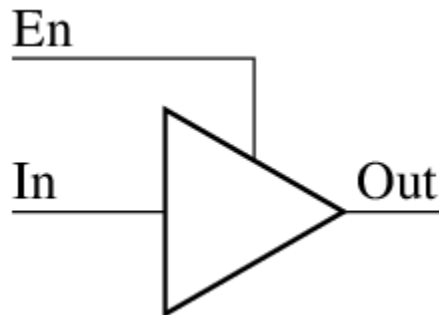


Day 24

TriState Buffer in Verilog

Circuit Diagram :



Enable	Input	Output
0	0	Hi-Z
0	1	Hi-Z
1	0	1
1	1	0

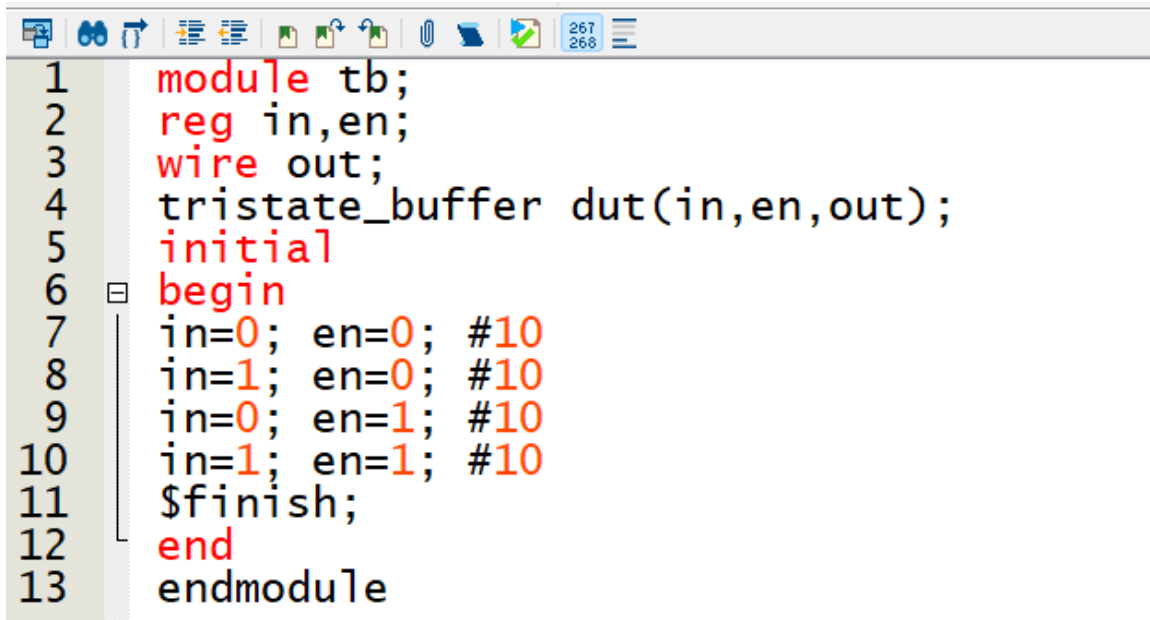
Verilog Code

Tristate buffer verilog code

```
1 module tristate_buffer(in,en,out);
2   input in,en;
3   output out;
4   assign out=en?in:1'bz;
5   endmodule
6
```

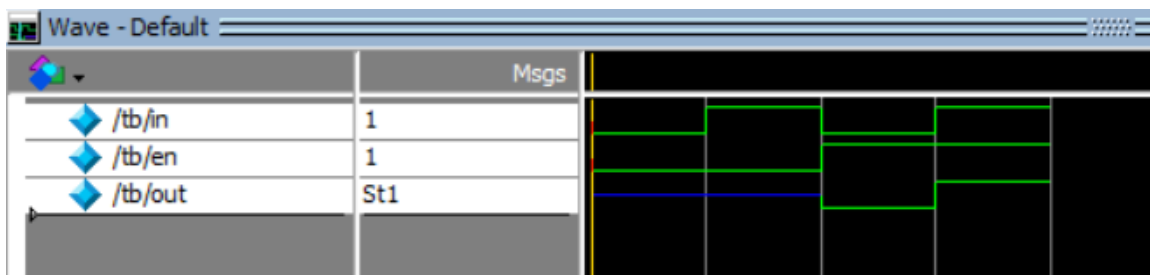
Testbench Code

Tristate buffer Tb code

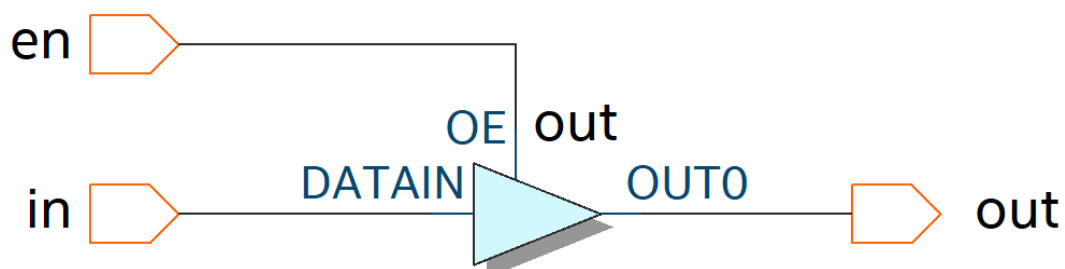


```
1  module tb;
2  reg in,en;
3  wire out;
4  tristate_buffer dut(in,en,out);
5  initial
6  begin
7      in=0; en=0; #10
8      in=1; en=0; #10
9      in=0; en=1; #10
10     in=1; en=1; #10
11     $finish;
12 end
13 endmodule
```

Waveform



Schematic





EDA Tools Used

- IntelQuartusPrime
- ModelSim

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